

A Partial Summary of 2020 USGS Activities of Interest to the FHWA and State Highway Agencies

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For questions, please contact either:

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Introduction and summary of current projects

The U.S. Geological Survey (USGS) serves the Nation by providing reliable scientific information to describe and understand the Earth; minimize loss of life and property from natural disasters; manage water, biological, energy, and mineral resources; and enhance and protect our quality of life. The USGS does much of this work in cooperation with State and Federal transportation and regulatory agencies as well as many land and water management agencies. As part of these efforts, the USGS has a long history of funded investigations with the Federal Highway Administration (FHWA) and State highway agencies to provide data and information to address various issues related to water resources and the Nation's transportation infrastructure. These issues cover a wide spectrum and include items such as regional flow statistics, flood documentation, regional stream characteristics, bridge scour, and water-quality assessments. One of the primary USGS products of interest to and DOTs and State highway agencies is the National Water Information System Web (<https://waterdata.usgs.gov/nwis>) and the related mapper (<https://maps.waterdata.usgs.gov/>), these tools provide current and historical data from more than 1.9 million sites across the country. These on-line databases provide data on the quality and quantity of surface water and groundwater. Another primary product is the USGS Publications Warehouse (<https://pubs.er.usgs.gov/>), which provides on-line open access to more than 160,000 publications written by USGS scientists since 1879. Many of these reports are applicable to one or more transportation research need.

The following table and text provide a partial summary of current or recently completed USGS activities related to highway issues. Table 1 lists the type of research by generalized subject and provides a list of States and (or) Federal agencies that are helping sponsor scientific investigations on each subject. The text following Table 1 provides more information about activities that have been or are being conducted on a national level and is followed by a partial list of reports published in 2020 that may be of interest to the transportation community. If you should have questions regarding this information, please contact Julie Kiang (jkiang@usgs.gov), and/or Gregory Granato (ggranato@usgs.gov).

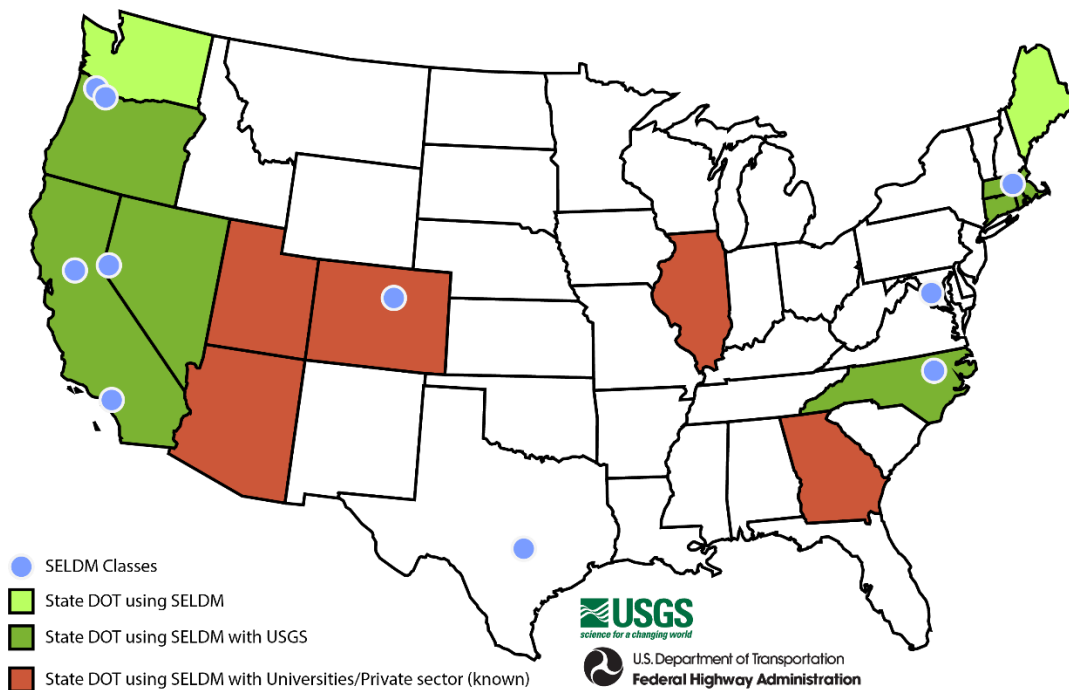
Table 1. Partial summary of active USGS projects with the FHWA and State Highway Agencies

Subject	Agency (FHWA or State DOT)
Ecology	
Flood frequency	FHWA, CO, GA, MS, ND, NM, SD, TX
Groundwater	LA
Hazards	FHWA, AL, CO, GA, MS, MO, MT, NM, SD, TX
Hydraulics	MO, NM, NC
Peak flow	FHWA, CO, GA, MS, NM, SD, TX
Scour	FHWA, CO, ID, MS, MO, MT, SC, TX, WA
SELDM	FHWA, CA, CT, MA, NV, OR, RI
Streamflow	AK, AL, AR, AZ, CO, GA, HI, IA, IN, LA, MI, MN, MO, MS, MT, NC, NJ, NM, NV, NY, OR, SC, SD, TN, TX, VA, VT, WA, WV
StreamStats	CO, CT, IN, LA, MA, RI, SC
Water Quality	FHWA, CT, IN, LA, MA, NC, NY, NV, OR, RI, WA, WV

Partial Summary of USGS National Activities

Training and technical support for the FHWA-USGS Stochastic Empirical Loading and Dilution Model (SELDM)

The purpose of the project is to help State DOTs and decision makers adopt and use SELDM by providing training and technical support for the model. The USGS, in cooperation with the FHWA, has developed and delivered a 3-day classroom training course, an abbreviated half-day training class, and a series of webinars. To date, more than 200 people from across the Nation who work for State DOTs, USEPA, USGS, and other agencies have attended the 3-day classes. This year, because of COVID-19, we did a blended synchronous and asynchronous on-line class. This FHWA National Technical Support Project also is helping SELDM studies in CA, CT, MA, NC, NV, RI and UT. This year we also published 6 scientific products.



Map showing known SELDM applications

2020 Publications:

Granato, G.E., Spaetzel, A.B., and Medalie, L., 2021, Statistical methods for simulating structural stormwater runoff best management practices (BMPs) with the Stochastic Empirical Loading and Dilution Model (SELDM): U.S. Geological Survey Scientific Investigations Report 2020–5136, 41 p., <https://doi.org/10.3133/sir20205136>.

Jeznach, L.C., and Granato, G.E., 2020, Comparison of SELDM simulated total-phosphorus concentrations with ecological impervious-area criteria: Journal of Environmental Engineering: v. 146, No. 8, 10 p. [https://doi.org/10.1061/\(ASCE\)EE.1943-7870.0001763](https://doi.org/10.1061/(ASCE)EE.1943-7870.0001763)

Granato, G.E., 2021, Best management practices statistical estimator (BMPSE) version 1.2.0: U.S. Geological Survey software release, <https://doi.org/10.5066/P9XBPIOB>.

Granato, G.E., and Jeznach, L.C., 2020, Model archive for analysis of the effects of impervious cover on receiving-water quality with the Stochastic Empirical Loading Dilution Model (SELDL): U.S. Geological Survey data release, <https://doi.org/10.5066/P9K0Y7XR>.

Granato, G.E., Medalie, L., and Spaetzel, A.B., 2021, Statistics for simulating structural stormwater runoff best management practices (BMPs) with the Stochastic Empirical Loading and Dilution Model (SELDL): U.S. Geological Survey data release, <https://doi.org/10.5066/P9X3ECTD>.

Spaetzel, A.B., Steeves, P.A., and Granato, G.E., 2020, Basin characteristics and point locations of road crossings in Connecticut, Massachusetts, and Rhode Island for highway-runoff mitigation analyses using the Stochastic Empirical Loading and Dilution Model: U.S. Geological Survey data release, <https://doi.org/10.5066/P9VK1MCG>.

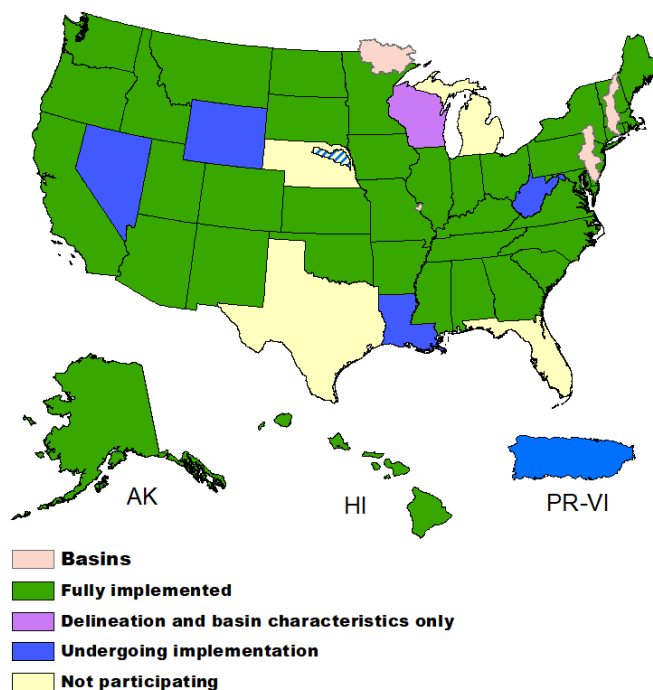
Many training modules are available on-line at

https://www.youtube.com/channel/UChgAb_28TbrqXUu8ZpombXw

Please contact Gregory Granato (ggranato@usgs.gov) for more information. For links to the published software, reports, and training information, please see the project web page <https://www.usgs.gov/SELDL> (DOI <https://doi.org/10.5066/F7TT4P3G>)

StreamStats Program

StreamStats (<https://streamstats.usgs.gov/ss/>) is a Geographic Information Systems-based Web application that greatly reduces the time needed for users to obtain streamflow statistics, basin characteristics, and other information for USGS data-collection stations and for ungaged sites. Users can select data-collection station locations shown on a map interface in a Web browser window to obtain previously published information for the stations. Users also can select any location along a stream to obtain the drainage-basin boundary, basin and climatic characteristics, and estimated streamflow statistics for the location. The estimates for ungaged sites are determined from USGS regional-regression equations and usually can be obtained in only a few minutes. Many hydrologic studies featured in the StreamStats application were done in cooperation with State DOTs. As of December 2020, StreamStats is available to the public for 42 states with full functionality. Currently (2020) StreamStats applications are undergoing implementation in Louisiana, Nevada, Puerto Rico, West Virginia, Wyoming. Applications also are available for two states and four river basins (Connecticut River, Delaware River Basin, Platte River, and Rainy River.) In 2020, the most relevant development in StreamStats was the addition of road miles and road crossings above any point in southern New England, this effort was completed in cooperation with the FHWA and CT, MA, and RI DOTs (Spaetzel and others, 2020).



Status of StreamStats implementation on December 31, 2020

Spaetzel, A.B., Steeves, P.A., and Granato, G.E., 2020, Basin characteristics and point locations of road crossings in Connecticut, Massachusetts, and Rhode Island for highway-runoff mitigation analyses using the Stochastic Empirical Loading and Dilution Model: U.S. Geological Survey data release, <https://doi.org/10.5066/P9VK1MCG>

If you would like to learn more about StreamStats, please contact Peter McCarthy (pmccarth@usgs.gov).

Updated Flood-Frequency Analysis Guidelines and Skew Maps

Flood-frequency analysis provides information about the magnitude and frequency of flood discharges. New guidelines were published in 2019 as Guidelines for Determining Flood Flow Frequency, Bulletin 17C as a product of a team organized under the Advisory Committee on Water Information's Subcommittee on Hydrology, Hydrologic Frequency Analysis Workgroup. The report defines procedures that provide a uniform and consistent approach for determining flood-flow frequency from peak-flow records. Improvements include more detailed representations of flood data, an improved computational algorithm, better screening for statistical outliers, improved methods for incorporation of regional information, and improved accuracy of confidence limits. The enhancements combine to provide more robust estimation of flood frequency. The updated methods are particularly valuable when new regional information is available or when historical flood records or paleoflood information are available. Note that while the estimation methods have been improved, the framework for flood frequency analysis has not been changed. As a result, Bulletin 17C methods allow continuity with previous studies. Bulletin 17C recommends weighting the at-site skew with a regional skew. Bulletin 17C has been published as a USGS Techniques and Methods report.

The USGS is applying the guidelines from Bulletin 17C in all new flood frequency studies including updates to station flood frequency estimates, as well as regionalization studies. The USGS is also working with FEMA and various state and local agencies to update the National flood-frequency skew map provided with Bulletin 17B.

Publications:

England, J.F., Jr., Cohn, T.A., Faber, B.A., Stedinger, J.R., Thomas, W.O., Jr., Veilleux, A.G., Kiang, J.E., and Mason, R.R., Jr., 2019, Guidelines for determining flood flow frequency—Bulletin 17C (ver. 1.1, May 2019): U.S. Geological Survey Techniques and Methods, book 4, chap. B5, 148 p., <https://doi.org/10.3133/tm4B5>.

The new guidelines and skew maps can be obtained at:

<https://acwi.gov/hydrology/Frequency/b17c/>.

The Peak flow FreQuency analysis program (PeakFQ) implements the Bulletin 17C recommended procedures for flood-frequency analysis of streamflow records.

Veilleux, A.G., Cohn, T.A., Flynn, K.M., Mason, R.R., Jr., and Hummel, P.R., 2014, Estimating magnitude and frequency of floods using the PeakFQ 7.0 program: U.S. Geological Survey Fact Sheet 2013–3108, 2 p., <https://dx.doi.org/10.3133/fs20133108>.

A Windows version of PeakFQ is available at <http://water.usgs.gov/software/PeakFQ/>.

Please contact Julie Kiang (jkang@USGS.gov) for more information about the USGS contributions to the effort.

Non-Stationarity in Flood Frequency Analyses

Flood frequency estimates are a major consideration in hydrologic design of culverts, bridges, and other infrastructure. In using historically observed peak flows to determine the peak design flow, the standard methods assume that the peak design flow will occur in the future with the same probability and magnitude as determined from the historical analysis. With changes to climate, land cover, snowpack, and agricultural and land drainage practices occurring across the United States and potential future changes, the assumption of stationarity in the observed peak flow record may not be valid.

Potential changes to floods may be the result of many factors, some of which operate at the watershed scale (such as changes to land-drainage or urbanization), some of which operate at the regional scale (such as changes to snowpack) and some of which operate at the global scale (such as changes to climate and large-scale weather patterns). The USGS, in cooperation with the FHWA, is using its extensive database of streamflow information and its site-specific knowledge about individual gages to advance the state of knowledge in three gap areas:

Characterize changes in peak flows: In this study, we extended on previous analyses by using data from long-record stream gages, to study observed trends - whatever the cause. Results are described in the three publications listed below. Selected results are being incorporated into an online viewer, to be released in early 2021. **Diagnose and attribute changes in peak flows:** To the degree possible, this study is attributing the observed flood trends to changes in both climate-based metrics as well as land-cover characteristics such as artificial land drainage and urbanization and other land cover change. A USGS report is expected later in 2021.

Adjusting flood-frequency analysis for observed and projected change: The study will include an assessment of different methods for incorporating trends into flood frequency analysis. A paper describing the analysis is in preparation.

Publications:

Publications related to diagnosing and attributing changes in peak streamflow include an eight-chapter USGS Professional Paper that has been reviewed and is currently in the USGS editing process and the following journal article.

Blum, A.G., Ferraro, P.J., Archfield, S.A., and Ryberg, K.R., 2020, Causal effect of impervious cover on annual flood magnitude for the United States: Geophysical Research Letters, v. 47, no. 5, p. e2019GL086480, <https://doi.org/10.1029/2019GL086480>.

Please contact Julie Kiang (jkiang@usgs.gov) or Karen R Ryberg (kryberg@usgs.gov) for more information.

Hydrologic Properties Affecting Scour and Channel Change

Historically, hydraulic engineers have used the peak flow or (occasionally) the rate of flow versus time (hydrograph) in designing hydraulic structures and modeling potential scour and channel change. Typical hydraulic practice estimates the peak flow or hydrograph from historical stream gage data or, for ungaged streams, deterministic and statistical methods (usually based on historical information). However, the new hydraulic methodologies require knowledge of additional hydrologic properties to advance and implement the methodologies. The USGS and FHWA are collaborating to enhance understanding of hydrologic properties and the potential future changes to hydrologic properties and to develop methods to characterize these hydrologic properties and allow their application to those situations such as scour and channel formation. For example, how would practice move beyond using peak flow; perhaps looking at how to represent a series of flow events over some service life? How could we do so where no gaging station information readily exists? This project is in its planning stages.

Please contact Julie Kiang (jkiang@usgs.gov) for more information.

Performance and Effectiveness of Scour Countermeasures

Scour countermeasures have become a major part of the FHWA's national bridge scour program and are considered vital in reducing the vulnerability of bridges to scour. However, lack of field verification of the performance and effectiveness of these countermeasures leaves uncertainty in the reliability of scour countermeasures for protecting foundations. FHWA, therefore, has teamed with the USGS to complete a comprehensive, national investigation of scour countermeasures. Through this investigation, FHWA hopes to evaluate and improve its published guidance and technical procedures for the selection, design, construction and maintenance of scour countermeasures and possibly reevaluate its policy of not using scour countermeasures at new bridge piers.

For this project, which was recently completed, the USGS has performed various levels of site evaluations at approximately 34 bridges with scour countermeasures across the Nation; USGS and FHWA teams have assessed bridges in Florida, Missouri, Idaho, Iowa, Missouri/Illinois (border bridge), Indiana/Illinois (border bridge), Missouri/Tennessee (border bridge), South Carolina, Iowa, Montana, New Jersey, Pennsylvania, and Connecticut. Some of the techniques used included stream-side investigations and underwater reconnaissance using state-of-the-art survey techniques such as terrestrial LiDAR, multi-beam bathymetry and side-scan sonar. At a small subset of these sites, USGS and FHWA personnel returned and actively pushed the limits of technology to collect additional very-high resolution sonar data that has not been readily available until this time. FHWA personnel worked with USGS to identify bridge locations and provide technical assistance at all levels; USGS scientists conducted the evaluations, documented each evaluation in templates, made all data and approved metadata available to FHWA via the web, and summarized data and findings in a series of official USGS reports (2 open-file reports and a final scientific-investigations report).

Publications:

Dudunake, T.J., Huizinga, R.J., and Fosness, R.L., 2017, Bridge scour countermeasure assessments at select bridges in the United States, 2014–16 (ver. 1.1, October 2017): U.S. Geological Survey Open- File Report 2017-1048, 10 p., <https://doi.org/10.3133/ofr20171048>.

Dudunake, T.J., Huizinga, R.J., and Fosness, R.L., 2019, Bridge scour countermeasure assessments at select bridges in the United States, 2016–18: U.S. Geological Survey Open-File Report 2019-1008, 12 p., <https://doi.org/10.3133/ofr20191008>.

Suro, T.P., Huizinga, R.J., Fosness, R.L., and Dudunake, T.J., 2020, Assessment of bridge scour countermeasures at selected bridges in the United States, 2014–18: U.S. Geological Survey Scientific Investigations Report 2019–5080, 29 p., <https://doi.org/10.3133/sir20195080>.

As a complement to the USGS site evaluations, the J. Sterling Jones Hydraulics Research Laboratory (HRL) continues to run hydraulic physical models and computational fluid dynamics (CFD) on several bridge sites to test and model the stability and performance of the as-built countermeasures observed in the field at high flows. These results will also be used to evaluate FHWA guidance on scour countermeasure design.

Please contact Peter Cinotto (pcinotto@usgs.gov) for more information.

National Streamflow Statistics

The National Streamflow Statistics (NSS) Program is a Microsoft Windows-based computer program created by the USGS to present high and low streamflow estimation equations for ungaged sites across the United States. Equations and their solutions in NSS provide low-flow duration and frequency estimates in addition to flood-frequency estimates such as the 100-year flood. Regression equations for estimating flood-frequency statistics of peak flows for rural and naturally flowing rivers are available for all 50 United States as well as the Commonwealth of Puerto Rico and the island of Tutuila, American Samoa. State-specific regression equations for estimating flood-frequency statistics of peak flows for urban streams are available in NSS for 18 U.S. States. Please contact Todd Koenig (tkoenig@usgs.gov) or Peter McCarthy (pmccarth@usgs.gov) for more information.

USGS WaterAlert and WaterNow

The USGS continues to provide a very popular water-threshold exceedance notification program. The system sends email or text messages when water levels, water-quality conditions, or rainfall meet user- specified criteria at real-time USGS hydrologic data collection sites. Criteria can include greater-than, less-than, within, and out-of-range thresholds. Reporting frequencies can include once-per-day or once- per-hour alerting while the condition lasts. The USGS WaterAlert system can be accessed at <https://maps.waterdata.usgs.gov/mapper/wateralert/>. A complementary interactive USGS query and alert feature called WaterNow is also available for users to query any real-time USGS hydrologic data collection site and request reports of the most recent values for any data collected at the site of interest. The query and response can be sent and received using any device with email or text message capabilities. Information about the USGS WaterNow system is available at: <http://water.usgs.gov/waternow/>

Flood Response Activities

The USGS, provides hydrologic information prior to, during, and after flooding, with a particular expectation that USGS will disseminate near real-time flood data and flood summaries on the World Wide Web. During major floods, the USGS deploys field crews to measure flood flows at streamgages and other locations of interest (such as breached levees, chemical spills, etc.). Flood-related data are available at the following locations:

<https://waterdata.usgs.gov>: time series stage, streamflow, and water quality data and at-site measurements

<https://stn.wim.usgs.gov/FEV/>: a national event-based tool for viewing and downloading storm tides sensor data and high-water mark data

<https://waterwatch.usgs.gov>: situational awareness maps and tools

<https://water.usgs.gov/floods>: a national website on major flooding events

Please contact Karl Winters (kwinters@usgs.gov), the USGS National Flood Hazard Coordinator, for more information.

Groundwater and Streamflow Information Program (GWSIP)

The U.S. Geological Survey (USGS) operates more than 8,200 near real-time streamgages in the National Streamflow Network (NSN) in cooperation with Federal, State, local, regional, tribal, non-governmental and industrial partners. The USGS provides a continuous source of streamflow information that is used in countless ways by governmental organizations, private industries, and the general public (<https://www.usgs.gov/gwsip>). Please contact Chad Wagner (cwagner@usgs.gov) for more information.

USGS Surface Velocity Workgroup

The Surface Velocity Work Group (“Surf Board”) is to assist the USGS Water Mission Area (WMA) in its overall mission to identify and develop new, advanced streamflow and velocity measurement and monitoring methods, provide guidance and training to most effectively deploy and use them, and to evaluate the trade-offs in the quality of the resulting data.

Please contact Frank Engel (fengel@usgs.gov) for more information.

Selected USGS publications from 2020

In 2020, the USGS produced 2,089 publications that are available on line from the USGS Publications Warehouse (<https://pubs.er.usgs.gov/>). Open access to more than 160,000 publications written by USGS scientists since 1879 are available on-line at the Publications Warehouse. Many of these reports may contain information useful for DOT decisionmakers. For example the search-link “ <https://pubs.er.usgs.gov/search?q=highway> ” yields a list of almost than 1,400 potentially relevant publications and the search link “<https://pubs.er.usgs.gov/search?q=scour> ” yields a list of more than 1,000 potentially relevant publications.

Ecology

Aquatic invasive species in the Chesapeake Bay drainage—Research-based needs and priorities of U.S.

Geological Survey partners and collaborators <https://pubs.er.usgs.gov/publication/ofr20201057>

Connectivity in the Crown: Highway 2 wildlife crossings <https://pubs.er.usgs.gov/publication/70210831>

Effects of box culverts on stream habitat, channel morphology, and fish and macroinvertebrate communities at selected sites in South Carolina, 2016–18 <https://pubs.er.usgs.gov/publication/sir20205021>

Historical changes in fish communities in urban streams of the southeastern U.S. and the relative importance of water-quality stressors <https://pubs.er.usgs.gov/publication/70206074>

Hydrologic modeling for flow-ecology science in the Southeastern United States and Puerto Rico
<https://www.srs.fs.usda.gov/pubs/59109>

Identification of management thresholds of urban development in support of aquatic biodiversity conservation
<https://pubs.er.usgs.gov/publication/70208347>

Integrating airborne remote sensing and field campaigns for ecology and Earth system science
<https://doi.org/10.1111/2041-210X.13463>

Isolating anthropogenic wetland loss by concurrently tracking inundation and land cover disturbance across the Mid-Atlantic Region, U.S. <https://pubs.er.usgs.gov/publication/70211980>

Sampling and analysis frameworks for inference in ecology <https://doi.org/10.1111/2041-210X.13279>

The use of Bayesian priors in Ecology: The good, the bad and the not great
<https://doi.org/10.1111/2041-210X.13407>

Urban stream syndrome and contaminant uptake in salamanders of Central Texas
<https://pubs.er.usgs.gov/publication/70210902>

Floods, Hazards, & Peak Flows

Causal effect of impervious cover on annual flood magnitude for the United States

<https://pubs.er.usgs.gov/publication/70217011>

Change points in annual peak streamflows: Method comparisons and historical change points in the United States <https://pubs.er.usgs.gov/publication/70209553>

Changes in event-based streamflow magnitude and timing after suburban development with infiltration-based stormwater management <https://pubs.er.usgs.gov/publication/70205935>

Channel narrowing by inset floodplain formation of the lower Green River in the Canyonlands region, Utah <https://pubs.er.usgs.gov/publication/70211926>

Characterization of peak streamflow and stages at selected streamgages in eastern and northeastern Oklahoma from the May to June 2019 flood event—With an emphasis on flood peaks downstream from dams and on tributaries to the Arkansas River <https://pubs.er.usgs.gov/publication/ofr20201090>

Deglacierization of a marginal basin and implications for outburst floods, Mendenhall Glacier, Alaska <https://pubs.er.usgs.gov/publication/70211973>

Delineation of flood-inundation areas in Grapevine Canyon near Scotty's Castle, Death Valley National Park, California <https://pubs.er.usgs.gov/publication/sir20205082>

Development of regional skew coefficients for selected flood durations in the Columbia River Basin, northwestern United States and British Columbia, Canada <https://pubs.er.usgs.gov/publication/sir20205073>

Does channel narrowing by floodplain growth necessarily indicate sediment surplus? Lessons from sediment-transport analyses in the Green and Colorado rivers, Canyonlands, Utah <https://pubs.er.usgs.gov/publication/70216016>

Estimating flood magnitude and frequency on streams and rivers in Connecticut, based on data through water year 2015 <https://pubs.er.usgs.gov/publication/sir20205054>

Estimating streamflow and base flow within the nontidal Chesapeake Bay riverine system <https://pubs.er.usgs.gov/publication/sir20205055>

Estimating flood magnitude and frequency on gaged and ungaged streams in Maine <https://pubs.er.usgs.gov/publication/sir20205092>

Estimating flood magnitude and frequency on streams and rivers in Connecticut, based on data through water year 2015 <https://pubs.er.usgs.gov/publication/sir20205065>

Flood-frequency estimation for very low annual exceedance probabilities using historical, paleoflood, and regional information with consideration of nonstationarity <https://pubs.er.usgs.gov/publication/sir20205065>

Flood-inundation maps for the Little Calumet River from Lansing to South Holland, Illinois, 2020 <https://pubs.er.usgs.gov/publication/sir20205074>

Flood-inundation maps for Dardenne Creek in St. Charles County, Missouri, 2019

<https://pubs.er.usgs.gov/publication/sir20205060>

Holocene paleofloods and their climatological context, Upper Colorado River Basin, USA

<https://pubs.er.usgs.gov/publication/70208533>

Hydrologic signals and surprises in U.S. streamflow records during urbanization

<https://pubs.er.usgs.gov/publication/70214485>

Magnitude and frequency of floods in Alabama, 2015 <https://pubs.er.usgs.gov/publication/sir20205032>

Precipitation, temperature, groundwater-level elevation, streamflow, and potential flood storage trends within the Brazos, Colorado, Big Cypress, Guadalupe, Neches, Sulphur, and Trinity River Basins in Texas Through 2017 <https://pubs.er.usgs.gov/publication/sir20195137>

Sea-level rise exponentially increases coastal flood frequency <https://pubs.er.usgs.gov/publication/70209476>

Steps to develop early warning systems and future scenarios of wave-driven flooding along coral reef-lined coasts <https://pubs.er.usgs.gov/publication/70209735>

Stormwater control impacts on runoff volume and peak flow: A meta-analysis of watershed modelling studies <https://pubs.er.usgs.gov/publication/70210513>

The roles of flood magnitude and duration in controlling channel width and complexity on the Green River in Canyonlands, Utah, USA <https://pubs.er.usgs.gov/publication/70216930>

Incipient bed-movement and flood-frequency analysis using hydrophones to estimate flushing flows on the upper Colorado River, Colorado, 2019 <https://pubs.er.usgs.gov/publication/sir20205069>

Predicting the floods that follow the flames <https://pubs.er.usgs.gov/publication/70210810>

InFRM Flood Decision Support Toolbox user guide Interagency Flood Risk Management (InFRM)

2020, Report <https://pubs.er.usgs.gov/publication/70209753>

Quantitative paleoflood hydrology <https://pubs.er.usgs.gov/publication/70211596>

Using small unmanned aircraft systems for measuring post-flood high-water marks and streambed elevations <https://pubs.er.usgs.gov/publication/70209903>

Hydraulics

Effects of box culverts on stream habitat, channel morphology, and fish and macroinvertebrate communities at selected sites in South Carolina, 2016–18 <https://pubs.er.usgs.gov/publication/sir20205021>

Hydrologic and hydraulic analyses of selected streams in Stark County, Ohio

<https://pubs.er.usgs.gov/publication/sir20205011>

Regional regression equations for estimation of four hydraulic properties of streams at approximate bankfull conditions for different ecoregions in Texas <https://pubs.er.usgs.gov/publication/sir20205086>

Landslide hydrology

Bellwether sites for evaluating changes in landslide frequency and magnitude in cryospheric mountainous terrain: A call for systematic, long-term observations to decipher the impact of climate change

<https://pubs.er.usgs.gov/publication/70215535>

Deep Learning as a tool to forecast hydrologic response for landslide-prone hillslopes

<https://pubs.er.usgs.gov/publication/70212531>

Four-dimensional surface motions of the Slumgullion landslide and quantification of hydrometeorological forcing <https://pubs.er.usgs.gov/publication/70211202>

Landslides across the United States: Occurrence, susceptibility, and data limitations

<https://pubs.er.usgs.gov/publication/70210102>

Landslide disparities, flume discoveries, and Oso despair <https://pubs.er.usgs.gov/publication/70213050>

Landslides after wildfire: Initiation, magnitude, and mobility <https://pubs.er.usgs.gov/publication/70212555>

Mobility characteristics of landslides triggered by Hurricane Maria in Puerto Rico

<https://pubs.er.usgs.gov/publication/70216279>

Rainfall triggers more deep-seated landslides than Cascadia earthquakes in the Oregon Coast Range, USA

<https://pubs.er.usgs.gov/publication/70214589>

Reconstructing the velocity and deformation of a rapid landslide using multiview video

<https://pubs.er.usgs.gov/publication/70212502>

The future of landslides' past—A framework for assessing consecutive landsliding systems

<https://pubs.er.usgs.gov/publication/70210033>

Twelve-year dynamics and rainfall thresholds for alternating creep and rapid movement of the Hooskanaden landslide from integrating InSAR, pixel offset tracking, and borehole and hydrological measurements

<https://pubs.er.usgs.gov/publication/70216486>

Scour

Assessment of bridge scour countermeasures at selected bridges in the United States, 2014–18

<https://pubs.er.usgs.gov/publication/sir20195080>

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